

Fluoridated Water: Old School Prevention or Still Essential?

Group 9

Abby Baker, Lauren Barger, Anslee Barnes, Marriette Brannan,
Laura Lally, Preston Little, Gracie Pollock, Walt Stewart, Arushi
Tandon

Faculty Mentor: Dr. Bachand



Behavioral Objectives

1. Describe the mechanism of action of fluoride in caries prevention, including its effects on enamel remineralization, demineralization resistance, and biofilm acid production.
2. Differentiate between evidence-based health benefits of community water fluoridation and the claims made in anti-fluoride movements by explaining the underlying science versus common misconceptions.
3. Evaluate the safety of fluoridated water by reviewing toxicological thresholds, recognized risks (fluorosis, acute ingestions), and the extensive scientific evidence supporting its safety for public health.
4. Compare and contrast the effectiveness of community water fluoridation with non-fluoride or no treatment alternatives, using guidelines and position statements from the ADA, CDC, and other public health organizations.
5. Apply patient centered communication strategies to address fluoride-related concerns in clinical practice, providing clear, evidence-based responses that support informed decision making.



BO 1. Describe the mechanism of action of fluoride in caries prevention, including its effects on enamel remineralization, demineralization resistance, and biofilm acid production.

Search Strategy

P: Individuals at risk for dental caries

I: Standardized fluoridation of water

C: No water fluoridation

O: Reduction in dental caries through
fluoride-induced remineralization and
prevention of demineralization

Databases used

Pubmed

Google Scholar

Keywords

Role of Fluoride in enamel
mineralization

Pubmed search string

(Fluoride) AND
(Remineralization)
(Fluoride) AND
(Demineralization)
(Fluoride) AND
(Biofilm Production)

Highest level of
evidence

Systematic Review



Mechanism of action of fluoride in caries prevention



Enamel Remineralization
Demineralization Resistance
Reduction of Biofilm acid
production



FLUORIDE PROMOTES REMINERALIZATION

Bacterial acid production demineralizes enamel

F- ions restore and promote the growth of Fluorapatite crystals in regions of partially demineralized enamel

Fluorapatite is stronger and more acid-resistant than hydroxyapatite

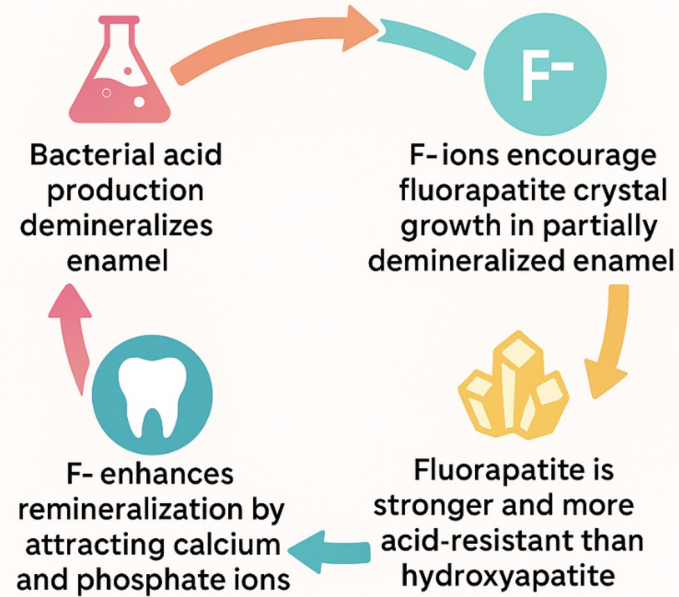
F- acts as a catalyst for remineralization by attracting calcium and phosphate ions

Via increasing rate of mineral deposition

F- promotes the formation of larger and more stable enamel crystals

Remineralized enamel + fluoride is less soluble and less vulnerable to future acid attacks

Fluoride Promotes Remineralization



Fluoride Inhibits Demineralization



Fluoride inhibits enamel demineralization.



When applied topically, fluoride contacts the enamel surface.



FLUORIDE INHIBITS BACTERIAL GROWTH AND ACID PRODUCTION IN PLAQUE BIOFILM

F- ions halt bacterial growth by inhibiting Enolase, but they need help entering the bacterial cell wall

In acidic environments (associated with bacterial activity), F- combines with H+ → HF

HF can surpass cariogenic bacterial cell wall, then dissociates into H+ and F-

F- ions prevent further acid production and bacterial energy production via inhibition of Enolase in *S. mutans*

F- ions now able to inhibit enolase activity

Enolase: Glycolysis enzyme in *S. mutans* that breaks down carbs and produces lactic acid





BO 2. Differentiate between evidence-based health benefits of community water fluoridation and the claims made in anti-fluoride movements by explaining the underlying science versus common misconceptions.

Search Strategy

P: Challenging effectiveness of water fluoridation (DMFT)

I: Fluoridation of water

C: No fluoridation of water

O: Reduction in caries risk as well as biological and physiological safety

Databases used

Pubmed

Google Scholar

NIH/CDC



Keywords

Effectiveness of fluoride in water

Pubmed search string

("Fluoridated water") AND ("IQ tests") AND ("systemic concerns")
("Fluoridated water") AND
("dental caries")

Search results

29 results, 21 results

Highest level of evidence

Systematic reviews, longitudinal studies

CLAIM: Fluoride causes cancer or bone fractures:
Long-term fluoride exposure can lead to osteosarcoma or brittle bones.

EVIDENCE

Early 1990 animal studies prompted concern, but **large scale epidemiologic studies show no consistent association.**

U.S. National Cancer Institute (NCI) & Public Health England reports found **no increased cancer risk** at fluoridated levels.

Bone fracture risk appears only at **very high natural fluoride concentrations (>4mg/L)**





CLAIM: Fluoride lowers IQ in Children: Even small fluoride doses impair neurodevelopment

EVIDENCE

Recent large-scale U.S. and Canadian studies show **no cognitive differences** at 0.7 mg/L.

Systematic reviews emphasize confounders (e.g., arsenic, socioeconomic status, nutrition).

Some studies in China and India link high fluoride exposure (>2 mg/L) to lower IQ, but these regions **far exceed** regulated fluoridation levels.



CLAIM: Fluoride accumulates in joints and causes arthritis.
Fluoride can suppress thyroid function.

EVIDENCE

No consistent link between **community-level exposure** and arthritis: pain reports often arise from **skeletal fluorosis** in high-fluoride regions (>4mg/L).

Thyroid studies (UK NHS, 2018; Malin et al., 2019) show **no suppression at recommended levels**, effects only appear in deficiency/overdose contexts.



Green et al., 2018

Prospective Canadian birth cohort
(~500 mother–child pairs).

Exposure measured via
spot urinary fluoride —
high day-to-day variability.

Found small IQ
reduction in boys
only, not girls.

Participants from both fluoridated & non-
fluoridated areas — **SES and education
differences** likely confounders.

Co-exposures (lead, arsenic,
etc.) not fully controlled.

Average exposure still
within normal North
American range; effect size
**within testing margin of
error.**

Authors caution that
findings **don't prove
causation** or apply to 0.7
mg/L community water.





Bashash et al., 2017

Longitudinal cohort in Mexican families
(≈ 300 mother–child pairs).

Fluoride sources: **salt fluoridation, tea, groundwater** — multiple unquantified exposures.

Average exposure 1.5–2.5 mg/L, well above regulated fluoridation levels.

Reported lower child IQ with higher maternal urinary fluoride.

Nutritional and socioeconomic factors (iodine, calcium, education) not fully adjusted.

Indicates risks only at **high, unregulated fluoride levels**, not at community standards.





CONSENSUS: *The dose makes the difference*

Regulated community fluoridation remains safe and beneficial.

Confounders like SES, nutrition, and co-exposures to other metals likely contributed.

Some longitudinal studies (e.g., MIREC, ELEMENT) observed associations between prenatal fluoride exposure and IQ — but at **higher-than-recommended doses**.

Large, high-quality cohorts (e.g., **Broadbent et al., 2015, New Zealand**) found **no IQ difference** at regulated fluoridation levels.





EVIDENCE BASED HEALTH BENEFITS

Initiation of water fluoridation results in reductions in:
DMFT of 1.81 (35%) and DMFT of 1.16 (26%)
Caries free children: ↑15%

CONCLUSION: (2015 SYSTEMATIC REVIEW)

Water fluoridation reduces caries levels in both deciduous and permanent dentition in children.
No conclusive evidence for effectiveness of water fluoridation for preventing caries in adults.

***DOWNSIDE:** Most studies are pre-1975 (71%)

An updated Cochrane review in 2024 has found that the dental health benefits of adding fluoride to drinking water may be lesser now -- since fluoride toothpaste is widely available.



ALASKA STUDY

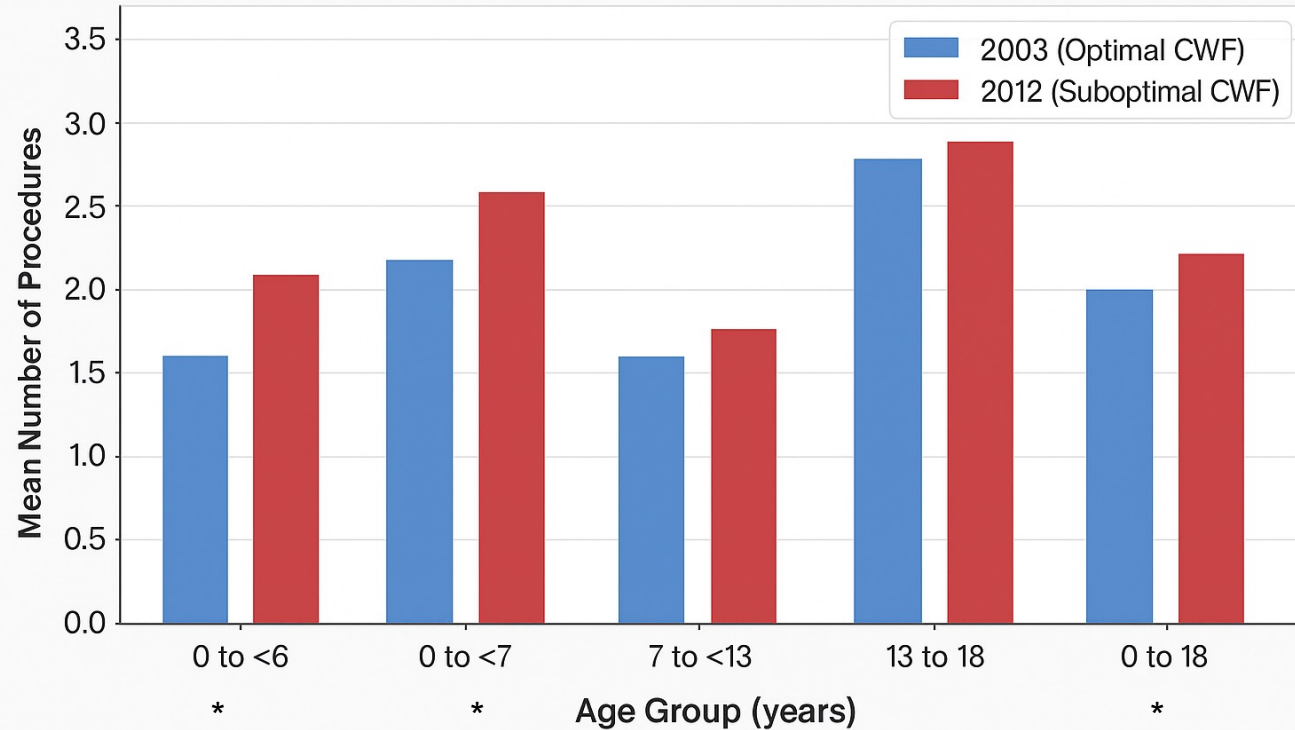
Juneau, Alaska (2003 vs. 2012) Retrospective study
CWF cessation in 2007: 96% residents on city water
~1000 Medicaid children (age 0-18), isolated area with relatively stagnant population
Widespread availability of fluoridated toothpaste throughout study

RESULTS

25.2% less chance for caries related procedure in 2003 vs. 2012 (51% age 0-6)
Caries treatment cost increased 73% for age <7



Mean Number of Caries Procedures per Child (2003 vs 2012)



*Statistical significance





BO 3. Evaluate the safety of fluoridated water by reviewing toxicological thresholds, recognized risks (fluorosis, acute ingestions), and the extensive scientific evidence supporting its safety for public health.

Search Strategy



PICO: Not Applicable

Databases used

Pubmed

Google Scholar

Trip Database

Galileo

Keywords

Fluorosis, Fluoride Toxicity, Fluoride thresholds, Acute Toxicity, Safety

Pubmed search string

((Fluoride) AND (Toxicity)) AND (Safety)
((Fluoride) AND (Adverse)) AND (Safety)

Search results

248 results

34 - reviews

5 - trials (randomized controlled, clinical)

Highest level of evidence

Systematic Reviews



Recommended Concentration of Fluoride Ions in drinking water = 0.7 ppm (0.7 mg/L) (CDC and US PHS)

RISKS:

Fluorosis (most important from birth-8 years)

Gastrointestinal Irritation and Discomfort

Bone Disease/Skeletal Fluorosis (pain and tenderness) (>4.0mg/L)

Death

40 L / 10.5 Gal
Of water
consumed to
reach acute
toxicity for 154
lb adult



56 24-oz
water bottles

ACUTE TOXICITY/INGESTION:

0.4-5 mg/kg b.w.

Nausea

Gastrointestinal Effects

Vomiting/Diarrhea

Lethal Dose = 40-80 mg/kg
b.w. (4000 L H₂O for 154
lbs adult)



FLUOROSIS: Most Common Complication

- Important during tooth development (Birth-8 y/o)
- Detectable change in enamel opacity (hypomineralization)
- 19% of 17 y/o patients (1986-1987)
- Of those, ~1.4% were classified >mild





Maximum Enforceable Fluoride
Concentration in Drinking Water (EPA)

4 mg/L

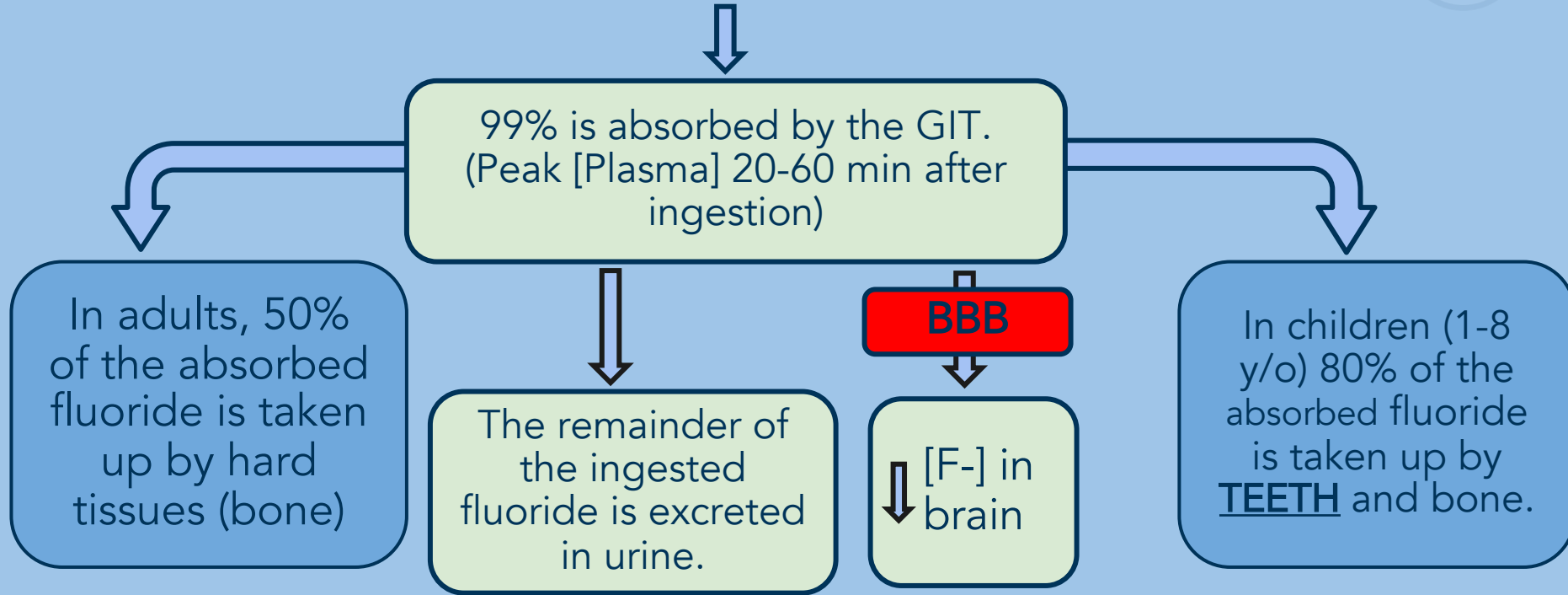
2 mg/L

The Secondary Standard of Fluoride Concentration in
Drinking Water (EPA)

Overall Recommended Fluoride Concentration in
Drinking Water (CDC)

0.7
mg/L

WHAT HAPPENS TO FLUORIDE AFTER INGESTION?



Evidence for Safety of Fluoride in our Water

Systematic Review from Zohoori, et al. (2025) includes studies from [2010 - present day] regarding nondental health outcomes from consumption of fluoridated water

THYROID FUNCTIONS

- Some studies found no correlation with fluoride and altered thyroid hormone production; other studies found evidence of changes in TSH, T3, and T4 concentrations at fluoride levels >1.5 mg/L
- No study accounted for factors that influence thyroid function such as micronutrient levels

CARCINOGENICITY

- No significant association was found between fluoride exposure and incidence of osteosarcoma or Ewing Sarcoma (even at elevated fluoride concentrations such as >2.0 mg/L)

SKELETAL QUALITY

- Evidence shows no significant harmful effects on adult skeletal health with consumption of low-concentration fluoridated water

IQ LEVELS

- Recent reviews only see possible cognitive neurodevelopmental impairment at fluoride concentrations well about 1.5 mg/L
- No evidence was seen at fluoride concentrations below 1.5 mg/L





OVERALL CONCLUSION FROM ZOOHORI, ET AL. (2025)

"...community water fluoridation at concentrations between 0.7 and 1.0 mg/L is not associated with adverse nondental health effects in the general population."

NOTE: This conclusion pertains to the nondental topics discussed as well as: cardiovascular outcomes, obesity outcomes, and metabolic outcomes (no adverse effects seen with 0.7-1.0 mg/L fluoridated water consumption)





BO 4. Compare and contrast the effectiveness of community water fluoridation with non-fluoride or no treatment alternatives, using guidelines and position statements from the ADA, CDC, and other public health organizations.

American Dental Association Official Position

Last updated in 2020



ADA endorses community water fluoridation as **safe, beneficial, cost-effective, and equitable**.

Supports fluoridation aligned with **U.S. Public Health Service** guidelines.

Encourages **dentist leadership** in initiating and maintaining programs.

Emphasizes maintaining **infrastructure and quality**.

Advises **education** about fluoride in bottled and filtered water.

Utah House Bill 81



- Fluoride ban to take place May 7, 2025
- Allows pharmacists to prescribe fluoride
- Reason: Individual choice, informed consent, and perceived health risk

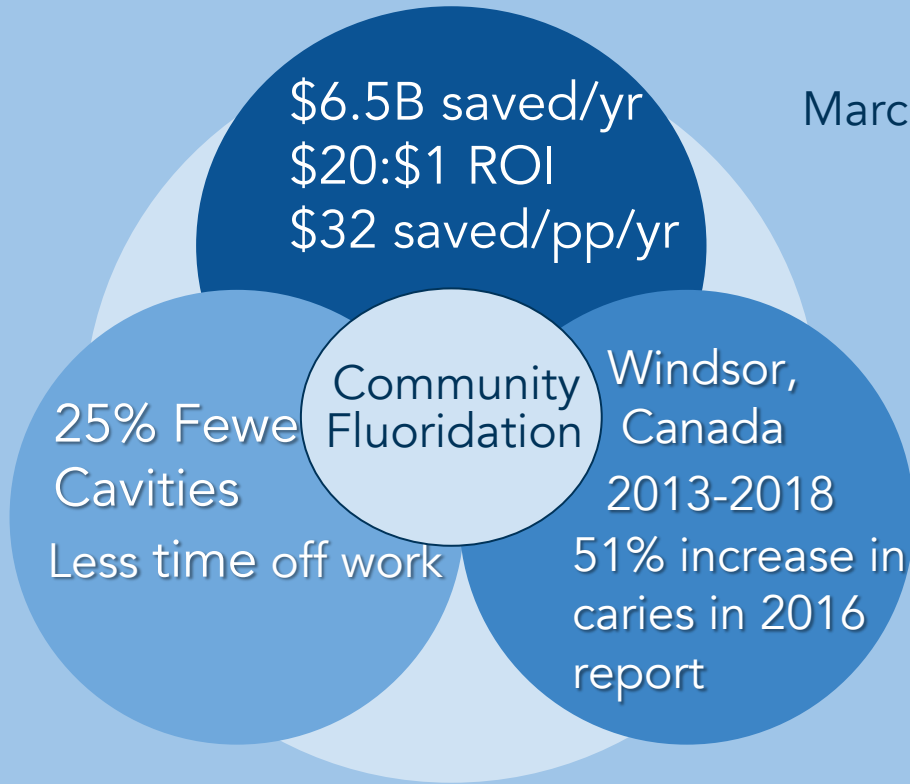


ADA PRESS RELEASE

ADA



March 28, 2025 in response to Utah Ban to take effect on May 7, 2025



"As a father and a dentist, it is disheartening to see that a proven, public health policy, which exists for the greater good of an entire community based on distorted pseudo-science"

Dr. Brett Kessler, ADA President



CDC



Named fluoridation 1 of the top 10 public health interventions of the 20th century

"It is a practical, cost effective, and equitable way to improve their resident's oral health regardless of age, education, or income."

0.7mg Fluoride/ L of water = **3** drops of water in **55** gallon barrel

NOTE: The CDC does not mandate the fluoridation of community drinking water



WORLD HEALTH ORGANIZATION (WHO)

<1.5ppm





BO 5. Apply patient-centered communication strategies to address fluoride-related concerns in clinical practice, providing clear, evidence-based responses that support informed decision-making.

Search Strategy

P: Patients who express concern, skepticism, or misinformation about fluoride use in dentistry

I: Patient-centered communication strategies

C: Standard clinician-centered communication

O: Improved patient understanding, trust, and willingness to accept fluoride-based preventive care; enhanced shared decision-making and patient satisfaction.

Databases used

Pubmed

Google Scholar



Keywords

Fluoride education, patient communication

Pubmed search string

("water fluoridation")
AND ("dentistry")AND
("communication")

Search results

11 found

Highest level of evidence

Systematic reviews, guidelines



FLUORIDE HESITANCY?

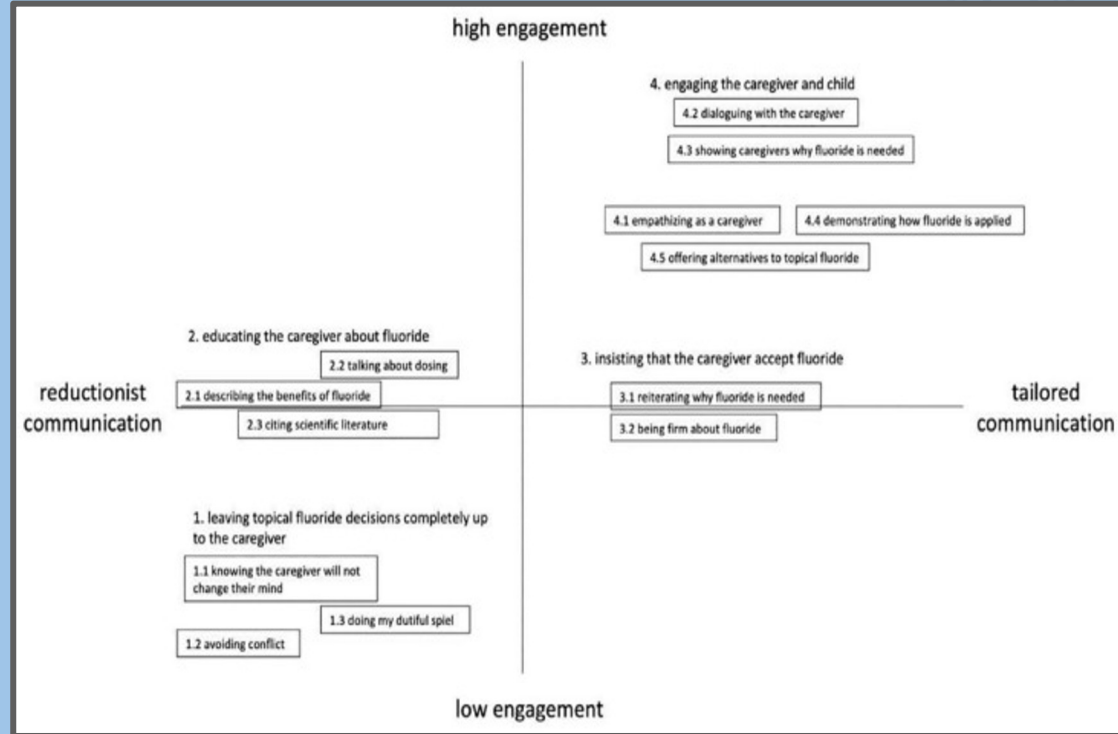
4 MAIN MODELS

Leaving topical fluoride decisions completely up to the caregiver.

Educating the caregiver about fluoride.

Insisting that the caregiver accept fluoride.

Engaging the caregiver and child.



PATIENT COMMUNICATION

Listen Attentively

Use Clear and Simple Language

Encourage Questions and Feedback

Use Visual Aids

Provide Reassurance and Empathy

Give Ample Time and Be Patient

Take a Patient-Centered Approach

Show Positive Body Language

Follow Up with Patients

Receive Continuing Education and Training



<https://texsmiles.com/services/>



WHAT DOES THE ADA HAVE TO SAY?

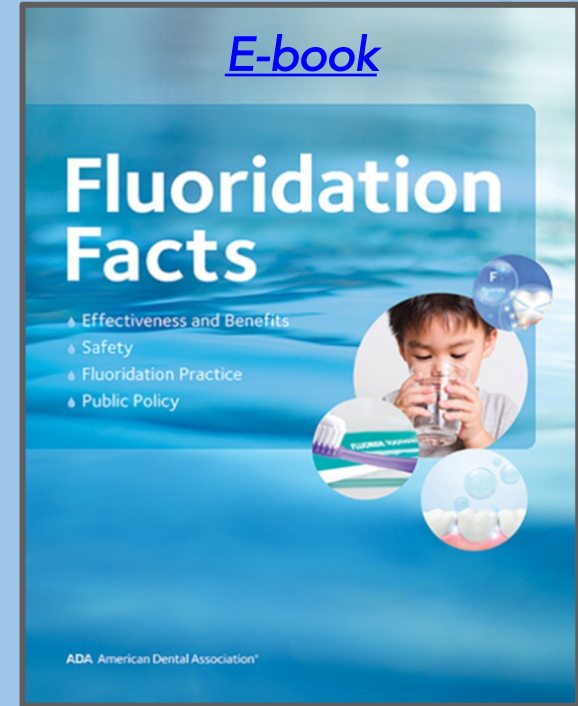


Remind patients that fluoride is *naturally occurring* and used widely

"START SMALL" APPROACH IS SUGGESTED: e.g., ask to use fluoridated toothpaste just at night or a couple times a week as a step.

CONSISTENCY - "I know you're not using fluoridated toothpaste... Let's see what we can do to prevent this."

PERSONAL - "...she wants what is best for their children and that she wouldn't suggest anything she isn't already doing in her own family."



PUT IT INTO PERSPECTIVE

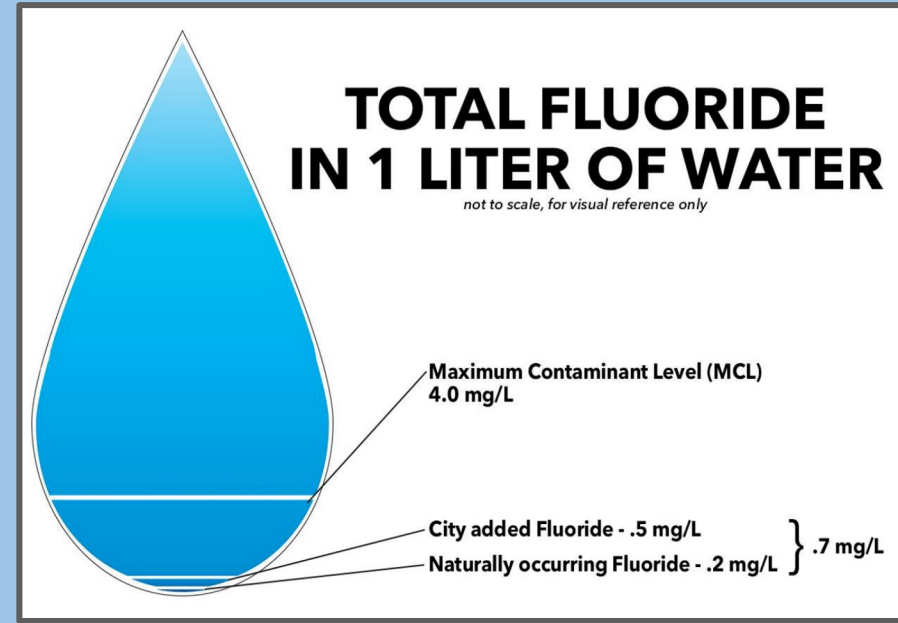
At 0.7 mg/L, there are seven-tenths of one part of fluoride mixed with 999,999.3 parts of water. While not exact, the following comparisons can be of assistance in comprehending 0.7 mg/L:

1 inch in approximately 23 miles

1 minute in approximately 1,000 days (2.73 yrs)

1 cent in approximately \$14,000

1 seat in more than 34 Wrigley Field baseball stadiums (seating capacity 41,268)



Guides to Patient Communication



"I can understand where you're coming from. Let me walk you through the scientific evidence."

Chairside Guide: Talking to Your Patients About Fluoride

Talking to Your Patients About Community Water Fluoridation

A Patient Engagement Guide for Dental & Medical Professionals

"Some of my other patients have asked a similar question. I'd be happy to share my expertise and answer any questions you may have..."

"It can be difficult to find unbiased information about this topic on the internet, so let me see if I can clarify some of what you've read..."

"I'm also a parent, so I understand why you might ask that question. Let me explain why this is so beneficial for your child(ren)..."



BO Review



1. Describe the mechanism of action of fluoride in caries prevention, including its effects on enamel remineralization, demineralization resistance, and biofilm acid production.

Fluoride aids in caries prevention by limiting demineralization, reducing bacterial acid production, and enhancing remineralization.

It does this by forming Fluorapatite crystals that are more resistant to acid erosion, aid in protecting against demineralization, and attract calcium and phosphate ions that also help remineralize compromised enamel.

BO Review



2. Differentiate between evidence-based health benefits of community water fluoridation and the claims made in anti-fluoride movements by explaining the underlying science versus common misconceptions.

Community water fluoridation results in a lower amount of decayed, missing, filled teeth for the deciduous and permanent dentition in children (dmft/DMFT).

Most studies that have been done have not accounted for interference from exposure to other metals and socioeconomic status.

Exposure to regulated levels of fluoride does not lead to systemic concerns.



BO Review



3. Evaluate the safety of fluoridated water by reviewing toxicological thresholds, recognized risks (fluorosis, acute ingestions), and the extensive scientific evidence supporting its safety for public health.

Acute Fluoride Toxicity/Ingestion: 0.4-5 mg/kg b.w.

Risks of Acute Fluoride Toxicity: Fluorosis (most important from birth-8 years), Gastrointestinal Irritation and Discomfort, Bone Disease/Skeletal Fluorosis (pain and tenderness) ($>4.0\text{mg/L}$), Death

Maximum Enforceable Fluoride Concentration in Drinking Water from EPA is 4 mg/L, and the overall recommended Fluoride Concentration from EPA is 0.7 mg/L.

Recent systematic review found fluoride levels between 0.7-1.0 mg/L in drinking water are not associated with adverse effects on nondental health.



BO Review



4. Compare and contrast the effectiveness of community water fluoridation with non-fluoride or no treatment alternatives, using guidelines and position statements from the ADA, CDC, and other public health organizations.

The ADA endorses community water fluoridation as **safe, beneficial, cost-effective, and equitable**.

Topical fluoride and other non-fluoride interventions are effective too but they depend on access, adherence, behavior change, etc. and therefore typically do not give the same broad public-health impact as CWF.

Major dental and public-health organizations (ADA, CDC, WHO) therefore maintain their endorsement of CWF as a cornerstone public-health strategy while also recommending complementary measures like topical fluoride and preventive care

BO Review



5. Apply patient centered communication strategies to address fluoride-related concerns in clinical practice, providing clear, evidence-based responses that support informed decision making.

4 Models

Leaving topical fluoride decisions completely up to the caregiver

Educating the caregiver about fluoride

Insisting that the caregiver accept fluoride

Engaging the caregiver and child.

What do YOU feel most comfortable with as a provider?

Listen Attentively

Use Clear and Simple Language

Put it into Perspective



References BO 1



- Buzalaf, M. A. R., Pessan, J. P., Honório, H. M., & Ten Cate, J. M. (2011). Mechanisms of action of fluoride for caries control. *Monographs in oral science*, 22, 97–114. <https://doi.org/10.1159/000325151>
- Iheozor-Ejiofor, Zipporah et al. "Water fluoridation for the prevention of dental caries." *The Cochrane database of systematic reviews* vol. 10,10 CD010856. 4 Oct. 2024, doi:10.1002/14651858.CD010856.pub3
- Malcangi, G., Patano, A., Morolla, R., De Santis, M., Piras, F., Settanni, V., Mancini, A., Di Venere, D., Inchingolo, F., Inchingolo, A. D., Dipalma, G., & Inchingolo, A. M. (2023). Analysis of Dental Enamel Remineralization: A Systematic Review of Technique Comparisons. *Bioengineering (Basel, Switzerland)*, 10(4), 472.
- Nassar Y, Brizuela M. The Role of Fluoride on Caries Prevention. [Updated 2023 Mar 19]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/sites/books/NBK587342/>



References BO 2



- Blakey, K. et al. (2014). *Osteosarcoma incidence and fluoridated water: Analysis of UK data*. *BMJ Open*, 4(8).
- Broadbent, J.M. et al. (2015). *Community water fluoridation and intelligence: Dunedin cohort study*. *Am J Public Health*, 105(1), 72–76.
- Choi, A.L. et al. (2015). *Fluoride exposure and bone health: A meta-analysis*. *Bone*, 74, 67–79.
- Iheozor-Ejiofor, et al. (2024). *Water fluoridation for the prevention of dental caries*. *Cochrane Database of Systematic Reviews*, 2024(11).
- Kim, H.N. et al. (2020). *Fluoride exposure and osteosarcoma risk: A systematic review and meta-analysis*. *Environmental Research*, 189, 109911.
- Malin, A.J. & Till, C. (2019). *Fluoride exposure and thyroid function: A systematic review*. *Environmental Health Perspectives*, 127(6), 067001.
- Meyer, J., Margaritis, V., & Mendelsohn, A. (2018). *Consequences of community water fluoridation cessation for Medicaid-eligible children and adolescents in Juneau, Alaska*. *BMC Oral Health*, 18(1).
- Muñoz, H.F. et al. (2023). *Fluoride exposure and cognitive function: Updated meta-analysis*. *NeuroToxicology*, 94, 111–120.
- Peckham, S. et al. (2018). *Fluoride and hypothyroidism: Assessment using UK health data*. *Journal of Epidemiology & Community Health*, 72(9), 865–870.
- Till, C. et al. (2020). *Fluoride exposure and neurodevelopment: A review*. *Environment International*, 134, 105225.
- Whitford, G.M. (2020). *Metabolism and health risks of fluoride: Review*. *Toxicology Letters*, 334, 85–93.



References BO 3 and 4



- American Dental Association. (2025, March 28). *Community water fluoridation prevents painful dental disease* [Press release]. <https://www.ada.org/about/press-releases/community-water-fluoridation-prevents-painful-dental-disease>
- American Dental Association. (n.d.). *Current policies: Fluoridation*. ADA. Retrieved November 11, 2025, from <https://www.ada.org/about/governance/current-policies#fluoridation>
- Anderson, K. (2025, February). Utah set to ban community water fluoridation. ADA News. American Dental Association. <https://adanews.ada.org/ada-news/2025/february/utah-set-to-ban-community-water-fluoridation/>
- Centers for Disease Control and Prevention. (2024, May 15). *Scientific statement on community water fluoridation: Statement on the evidence supporting the safety and effectiveness of community water fluoridation*. <https://www.cdc.gov/fluoridation/about/statement-on-the-evidence-supporting-the-safety-and-effectiveness-of-community-water-fluoridation.html>
- Fatemeh V Zohoori, Elizabeth A Kumah, Jelena Kronic, Michael Drinnan, Alexander J Morris, Public Health Impacts of Water Fluorides: Current Evidence from a Rapid Systematic Review, *Advances in Nutrition*, 2025, 100547, ISSN 2161-8313, <https://doi.org/10.1016/j.advnut.2025.100547>.
- World Health Organization. (2004). *Fluoride in drinking-water: Background document for development of WHO Guidelines for Drinking-water Quality (WHO/SDE/WSH/03.04/96)*. <https://www.who.int/docs/default-source/wash-documents/wash-chemicals/fluoride-background-document.pdf>



References BO 5



Anderson, O. (2025, June 30). *Community water fluoridation: Untangling facts from fear*. Olivia Anderson.
<https://adanews.ada.org/ada-news/2025/june/community-water-fluoridation-untangling-facts-from-fear/>

Arcora Foundation. (2024, May 14). *A patient engagement guide for dental & medical professionals* [PDF].
California Dental Association. https://www.cda.org/wp-content/uploads/Patient-Engagement-Guide_05.14.2024.pdf

Cruz, S., Ko, A., & Chi, D. L. (2024). A Qualitative Study on Dentists' Communication Approaches in Managing Fluoride-Hesitant Caregivers. *JDR Clinical & Translational Research*.
<https://doi.org/10.1177/23800844231203673>

Fluoridation facts, 2025 e-book page 11. (n.d.). <https://ebooks.ada.org/fluoridationfacts/21>

Ho JCY, Chai HH, Lo ECM, Huang MZ, Chu CH. Strategies for Effective Dentist-Patient Communication: A Literature Review. *Patient Prefer Adherence*. 2024;18:1385-1394. <https://doi.org/10.2147/PPA.S465221>

Michigan Department of Health & Human Services. (n.d.). *Chairside guide: Talking to your patients about fluoride* [PDF]. https://www.michigan.gov/-/media/Project/Websites/mdhhs/Folder3/Folder41/Folder2/Folder141/Folder1/Folder241/Chairside_Guide_Fluoride.pdf?rev=2dc391cbb5b34e83aa9c60916a9a21f5

New York State Department of Health. (n.d.). *Publication No. 0790* [PDF].
<https://www.health.state.ny.us/publications/0790.pdf>





Thank you!



Six Thinking Hats®



White Hat (Faculty): Provides objective data and scientific evidence about the dental material.



Red Hat (Patient): Expresses emotions, concerns, and preferences regarding the material.



Black Hat (Dentist): Evaluates potential risks, challenges, and clinical implications.



Yellow Hat (Sales Rep): Highlights the benefits and advantages of the material.



Green Hat (Manufacturer): Explores innovation, and future possibilities for the material.



Blue Hat (Board of Dentistry): Ensures all perspectives and standards are considered.





SKIT LINK



https://docs.google.com/document/d/1msqDr7IWpBHI8ZDE7qMnpB6-Z9K6_3fpckHK-rETo_g/edit?usp=sharing

